Vibrational Geometrodynamics VG – SERIES 5 Episode 2 :

INTRINSIC MASS AS A POSTULATIONAL DEFINITION :

The PMI Framework

Author: Physicist Pop Rodica & DeepSeek R1 Assistant

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ABSTRACT

We propose that mass is not generated by an external field, but is an intrinsic property of a stable nodal pattern — a soliton — whose coherence is maintained through pair fluctuations (virtual particles). The central relation is $\mathrm{m}_0 c^2 = \sum_i w_i \, \hbar\omega_i$, defining rest mass as the weighted sum of the nodal energies that constitute the pattern , where $w_i = \sum_j J_{ij}$ are scalar weighting coefficients. Within VG/RTS, the Higgs field does not create mass; it modulates the resistance to displacement of the pattern through the nodal medium. The postulate is grounded in bi-emergence (Instructions ↔ Vibrations), the operator interpretation of virtual particles, the nodal soliton structure, and phenomenological observations .

1. The central idea

Mass is not generated by an external field (Higgs), but is an intrinsic property of a stable nodal pattern — a soliton — which maintains its coherence through pair fluctuations (virtual particles).

This mass remains constant as long as the pattern preserves its integrity, regardless of the nodal medium through which it "swims". **The Higgs field does not create mass; it modulates the resistance to displacement of the pattern through the nodal field.**

2. Mathematical formulation

where:

$$m_0 c^2 = \sum_{i \in (pattern)} w_i \hbar \omega_i$$

- m_0= the intrinsic mass of the pattern (constant, if the pattern remains stable)
- ω_i = the nodal frequencies composing the pattern
- $w_i = \sum_j J_{ij}$ are scalar weighting coefficients constructed from the nodal coupling operators J_{ij} (cf. Bi-emergence chapter 4)

Note on the status of Equation (2.1).

Equation (2.1) is a postulational definition, not a derived result. We define intrinsic mass as the weighted sum of nodal energies; this is the foundational postulate of the PMI.

Degeneration condition.

When the weighting coefficients reduce to unity ($w_i \to 1$) , Equation (2.1) reduces to the simple sum:

$$m_0 c^2 = \sum_i \hbar \omega_i$$

3. Stabilization mechanism — The role of virtual particles

The nodal soliton (the stable pattern) is maintained through a field of pair fluctuations (virtual particles) that act as transition operators between:

· Intention($\mathcal{L}_{\text{vivo}}$)— the source of pattern updating

· Nodal imprint — the configuration archived in the disc

These fluctuations are not simple quantum noise. They are intersection points in the Network grid — analogous to the holes in an embroidery, through which the informational thread rises and falls to weave the pattern.

· When fluctuations are active and coherent, the soliton remains stable (constant mass).

· When fluctuations cease or become incoherent, the pattern unravels (mass dissipates).

4. Consequences for Series 5

1. **Biological identity** — a living organism is a stable nodal soliton, maintained by a field of pair fluctuations. Its intrinsic (informational) mass remains constant during its lifetime.

2. Imperfect reconstructions (Mediași) — when a pattern is reprojected from the archive, without its fluctuation field, it possesses no intrinsic mass. That is why the characters were mechanical and lifeless. It is a shadow of the pattern.

3. **DNA as nodal spiral** — DNA can be modeled as a stable soliton, whose pair fluctuations maintain informational coherence. This explains DNA repair and resonance effects.

4. **The cosmic archive** — the flattened disc does not store intrinsic masses, only nodal imprints (relational instructions). Intrinsic mass appears only when a pattern is reactivated through pair fluctuations.

5. **The Terra Program and PIGs** — a PIG (Genetic Engineering Program) can be understood as an experiment of maintaining or modifying nodal solitons (populations) by manipulating pair fluctuations (transition operators).

5. Connections with existing episodes

Concept Episode Relation to PMI

Re-anchoring (RTS25) - Pattern movement through resonance, without losing intrinsic mass

Interface operator (RTS29)- $\mathcal{I}$

modulates pair fluctuations, stabilizing or destabilizing the soliton

Bi-emergence (RTS30) - Instructions ↔ Vibrations mediated by pair fluctuations

Nodal cavity (VG4.2)- Frequency ω_i is the basic component of intrinsic mass

Topological Gates (VG4.7) - Nodal points are zones where pair fluctuations are more active, allowing pattern crossing

Relational Crystal (VG4.1) - State of perfect alignment — where fluctuations are absent, and intrinsic mass is undefined (pure potential)

6. Open questions for Series 5

1. How can the intrinsic mass of a C_{self}be measured?

2. Is there a relation between $m_0$and α (the constant in the Hubble equation)?

3. How does intrinsic mass behave in zones of extreme nodal density (black holes, inversion points)?

4. Can a C_{self}modify its own intrinsic mass through intention?

What's new?

The Postulate of Intrinsic Mass (PMI) proposes a paradigm shift: mass is not generated by an external field (Higgs), but is an intrinsic property of a stable nodal pattern – a soliton – maintained through pair fluctuations (virtual particles).

The central relation $m_0 c^2 = \sum_i w_i \hbar \omega_i$ expresses rest mass as the summed energy of the nodal modes that constitute the pattern. Within VG/RTS , the Higgs field does not create mass – it modulates the resistance to displacement of the pattern through the nodal medium. The postulate opens a new path for understanding biological identity, imperfect reconstructions (Medias), DNA as a nodal spiral, and the cosmic archive .

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